Cambridge Assessment International Education

Cambridge IGCSE™ (9–1)

CHEMISTRY (9–1) **0971/41**

Paper 4 Theory (Extended) **May/June 2022**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6	<u>Calculation specific guidance</u> Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'. For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values. For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme. Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.
7	<u>Guidance for chemical equations</u> Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme. State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)	carbon dioxide	**1**
1(b)	aluminium oxide	**1**
1(c)	ethanol (1) carbon dioxide (1)	**2**
1(d)	ethanol	**1**
1(e)	oxygen	**1**
1(f)	carbon dioxide	**1**
1(g)	nitrogen	**1**
1(h)	diamond	**1**

Question	Answer	Marks
2(a)(i)	charge: –1; 0; (1) relative mass: 1; 1; (1) **Mark by column**	**2**

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Question	Answer	Marks
2(a)(ii)	**12 | 12 | 12 (1)** **12 | 13 | 12 (1)** **Mark by row**	**2**
2(a)(iii)	(they have) 2 more protons than electrons **OR** (they have) 2 fewer electrons than protons **OR** (they have) 12 protons and 10 electrons	**1**
2(b)	Na^+ **or** Al^{3+} (1) F^- or O^{2-} or N^{3-} (1) Ne (1)	**3**

Question	Answer	Marks
3(a)(i)	positive ions / cations (1) sea of electrons / mobile electrons / delocalised electrons (1) attraction between positive ions and electrons (1)	**3**
3(a)(ii)	**electrons** move / **electrons** mobile / **electrons** flow	**1**
3(b)(i)	ionic	**1**

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Question	Answer	Marks
3(b)(ii)	**ions** move / **ions** are mobile / **ions** flow	**1**
3(c)(i)	pipette	**1**
3(c)(ii)	yellow to orange	**1**
3(c)(iii)	at least two results are within 0.2 cm^3 or less	**1**
3(c)(iv)	0.005 / 5×10^{-3} (1) 0.0025 / 2.5×10^{-3} (1) 0.125 (1)	**3**

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Question	Answer	Marks
4(a)(i)	$S + O_2 \rightarrow SO_2$	**1**
4(a)(ii)	(temperature) 450 °C (1) (pressure) 1–2 atmosphere(s) (1) vanadium(V) oxide catalyst (1) $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (1)	**4**
4(a)(iii)	$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$	**1**
4(a)(iv)	water	**1**
4(b)(i)	2 bonding pairs as **one dot and cross** each (1) 2 lone pairs on S (and no additional electrons on Hs) to complete the outer shell on S and both Hs (1)	**2**
4(b)(ii)	$\mathbf{2}H_2S + SO_2 \rightarrow \mathbf{3}S + \mathbf{2}H_2O$	**1**

Question	Answer	Marks
5(a)	the **rate** of forward reaction equals the rate of the reverse reaction (1) **concentrations** of reactants and products are constant (1)	**2**
5(b)	reaction too slow (1) yield of ethanoic acid too low (1)	**2**
5(c)	increases (1) / decreases (1) / decreases (1)	**3**

Question	Answer	Marks
5(d)	cobalt (1) transition element (1)	**2**
5(e)	$C_nH_{2n+1}COOH$	**1**
5(f)	COOH fully displayed (1) whole molecule completely correct (1)	**2**
5(g)(i)	methyl butanoate	**1**
5(g)(ii)	butanoic acid (1) methanol (1)	**2**

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Question	Answer	Marks
5(h)	C 48.65 / 12 H 8.11 / 1 O 43.24 / 16 **OR** 4.05:8.11:2.70 (1) fractions shown dividing all by smallest **OR** 1.5:3:1 **OR** 3:6:2 (1) $C_3H_6O_2$ (1)	**3**
5(i)	$C_4H_8O_2$	**1**

Question	Answer	Marks
6(a)(i)	zinc blende	**1**
6(a)(ii)	heat(zinc sulfide) strongly in air / roast in air	**1**
6(a)(iii)	carbon **or** carbon monoxide	**1**
6(a)(iv)	the temperature in the furnace is above **or** higher than the boiling point of zinc **OR** the boiling point of zinc is below **or** less than the temperature of the furnace	**1**
6(a)(v)	condensation / condensing / condense	**1**
6(b)(i)	no bubbles **or** no fizzing **or** no effervescence	**1**

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Question	Answer	Marks
6(b)(ii)	zinc / zinc oxide / zinc hydroxide	**1**
6(b)(iii)	(powder has) larg**er** surface area **OR** lumps have small**er** surface area (1) (powder has) more collisions per unit time / more collision frequency **OR** **lumps** have fewer collisions per unit time / less collision frequency (1)	**2**
6(b)(iv)	crystals (form on glass rod or microscope slide)	**1**
6(b)(v)	$ZnSO_4$	**1**

Question	Answer	Marks
7(a)(i)	from left to right caesium → rubidium→ potassium →sodium → lithium	**1**
7(a)(ii)	caesium hydroxide	**1**
7(b)	Group I element is less strong / not strong **ORA** **OR** Group I element has low(er) density **ORA** **OR** Group I element is soft(er) **ORA**	**1**
7(c)	solid	**1**
7(d)(i)	colourless (1) orange / brown / yellow (1)	**2**

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Question	Answer	Marks
7(d)(ii)	Br^- (1) loses electron(s) (1)	**2**
7(e)	432(1) 436(1) – 4(1)	**3**

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